Arista Access Node MALL-ACC-80 Routing Protocol Flap SOP (Minor Severity)

Node: MALL-ACC-80 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6827

Description

On Arista Access nodes, particularly MALL-ACC-80, routing protocol flaps are a common occurrence due to the dynamic nature of BGP, OSPF, and ISIS routing protocols. The flapping of routes can be attributed to various factors such as temporary network topology changes, misconfigured routing policies, or even hardware issues. In this specific scenario, the failure domain is centered around the Access node's role in the network, where the node is responsible for advertising routes to the core network. The protocol flap is triggered by the loss of OSPF adjacency, leading to a cascading effect on the ISIS LSP flooding, ultimately resulting in BGP route churn. This SOP aims to troubleshoot and resolve the issue by identifying the root cause and implementing corrective actions to stabilize the routing protocol and prevent future occurrences.

Pre-requisite Checks

- Pre-Requisite: Verify the Malleswaram Node's power and environmental conditions are within normal operating parameters., Description: This is to ensure that the node's hardware is not experiencing any issues due to power or environmental factors.

- Pre-Requisite: Check the redundant link status to ensure it is functioning correctly., Description: This is to verify that the redundant link is operational and can take over in case of a failure.

- Pre-Requisite: Gather logs and historical data to analyze the protocol flap pattern., Description: This will help in identifying any recurring patterns or anomalies in the protocol flap behavior.

Potential Root Causes

Cause	Probability
Temporary network topology changes causing OSPF adjacency loss	High
Misconfigured routing policies leading to BGP route churn	Medium
Hardware issues such as faulty interfaces or optics	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip ospf neighbor detail, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command will display the OSPF neighbor table, helping to identify if there are any issues with OSPF adjacency.

Command: show bgp summary, Expected Output: BGP router identifier 192.168.1.1, local AS number 64512, Description: This command will display the BGP routing table, helping to identify if there are any issues with BGP route churn.
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```
Command: show isis database, Expected Output: ISIS LSP Database (Sequence/LS ID/ Age), Description: This command will display the ISIS LSP database, helping to identify if there are any issues with ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify OSPF adjacency, Command: show ip ospf neighbor detail, Description: Verify OSPF adjacency is established and stable.

Step 2: Step: Check BGP routing table, Command: show bgp summary, Description: Verify BGP routing table is stable and consistent.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: Clear ISIS LSP database to reset the ISIS routing protocol.

Post-Check Verification

```
Command: show ip ospf neighbor detail, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify OSPF adjacency is stable and established.

Command: show bgp summary, Expected Output: BGP router identifier 192.168.1.1, local AS number 64512, Description: Verify BGP routing table is stable and consistent.

Command: show isis database, Expected Output: ISIS LSP Database (Sequence/LS ID/ Age), Description: Verify ISIS LSP database is stable and consistent.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy running-config startup-config, Description: Restore the original configuration to prevent any configuration changes from causing further issues.

- Step: Reset OSPF and BGP protocols, Command: clear ip ospf process, Description: Reset OSPF protocol to its original state.

- Step: Reset ISIS protocol, Command: clear isis database, Description: Reset ISIS protocol to its original state.